

Question number	Answer	Additional guidance	Mark
6(a)(i)	fast twitch (1)	accept type 2 / type II / type IIa / type IIb / type IIx	(1)

Question number	Answer	Additional guidance	Mark
6(a)(ii)	A description that includes three of the following points: - lactate diffuses (from muscle) into blood (1) {oxidation of lactate / lactate converted} to pyruvate (1) (and then) used in {link reaction / aerobic respiration} (1) (taken up by the liver where it is) converted to {glucose / glycogen} (1)	accept lactate goes to the liver accept {removal of hydrogen atoms / dehydrogenation} to convert to pyruvate ignore Krebs cycle accept stored as glycogen in the liver	(3)

Question number	Answer	Additional guidance	Mark
6(a)(iii)	• B is the correct answer A is not the correct answer as reduced capillary network is not correct C is not the correct answer as reduced capillary network and low myoglobin are not correct D is not the correct answer as low myoglobin is not correct		(1)

Question number	Answer	Additional guidance	Mark
6(a)(iv)	• D is the correct answer A is not the correct answer as Flick's law does not explain how muscles contract B is not the correct answer as fluid mosaic model does not explain how muscles contract C is not the correct answer as lock and key hypothesis does not explain how muscles contract		(1)

Question number	Answer	Mark
* 6(b)	Indicative content consideration of data in table •resting heart rate in athletes lower •peak heart rate higher in non athletes •maximum, volume of oxygen used higher in athletes in both aerobic and anaerobic conditions explanation of data in table • athletes have a lower resting heart rate as have greater {heart muscle / greater stroke volume / greater cardiac output} • non athletes peak heart rate is higher than athletes as heart needs to contract {more / faster} to get oxygen to tissues as {less developed heart muscle / lower peak flow} • athletes use more oxygen due to greater quantity of muscle cells (to be used in aerobic respiration) consideration of data in graph •as power generated increases oxygen used increases •athletes use more oxygen than non-athletes •athletes generate more power than non-athletes •in the graph non athletes have lower VO₂ at the same power explanation of data in graph • athletes use more oxygen • meaning that the metabolic efficiency / circulatory system efficiency athletes is higher • athletes have greater lung capacity • athletes have more muscles so can achieve greater power • resulting from years of training • athletes have more slow twitch fibres which give greater power • leading to increased muscle growth / development:/ increased lung capacity and gas exchange: increased metabolism /increased energy production in cells/ increased aerobic respiration/increased ATP production / more mitochondria / more myoglobin	

5	circulatory and respiratory system • circulatory and respiratory system work together to enable exercise • heart rate in athletes lower than non athletes compensates for by having an increase in strength of beats – bigger heart – more heart muscle. • increase in arterial pressure caused by increased cardiac output. • cardiac output = heart rate x stroke volume • ref to lung capacity / peak flow of athletes / ventilation rate	
6	muscles • training increases muscle growth / repair / replacement / capillary network • athletes muscle have greater capillary network • depending on exercise will increase amount of slow and fast twitch muscle fibres • more ATP can be used for breaking cross bridges / movement of calcium ions • contain more myoglobin to store oxygen • in non athletes muscles atrophy which affects the ability to do exercise and venous flow back to the heart	
7	biochemistry • more ATP due to more mitochondria / aerobic respiration / more oxygen available • athletes have greater tolerance to lactic acid from anaerobic respiration due to training • more oxygen carried to cells for aerobic respiration • BMR higher in athletes to release energy	
8	consideration of methodology • in table no SD / error bars / statistical test so no indication of validity or range of data • in graph no SD/error bars / statistical test so no indication of validity or range of data • comment on range of data in graph / overlap between athletes and non athletes • comment on methodology – age / gender / amount of training Level 1: Description of (a minimum of 2 specific) results from table and graph Level 2: Reference to all of level one plus one explanation of the effects of exercise / training with some details on physiological processes Level 3: Reference to all of level one and two plus one detailed explanation of the effects of exercise / training with significant details on physiological processes at cellular / genetic level.	(6)

